# A Multilingual Mechanistic Benchmark for Moral Decision-Making in LLMs

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FOSE7901 – STEM Research Placement 2

## Project Pitch

Large language models (LLMs) are increasingly used across languages and cultural contexts, yet semantically equivalent prompts can sometimes lead to different answers, confidence levels, or internal representations. Placement 1 established a practical foundation for investigating these differences by implementing sparse autoencoders (SAEs), the Jacobian Lens (JLens), and activation patching. SAEs provide a feature-level view of model activations (Huben et al., 2024), while JLens identifies intermediate representations associated with concepts that may later be verbalised (Gurnee et al., 2026). Previous research has also shown that multilingual models can contain both shared and language-specific internal representations (Deng et al., 2025; Dumas et al., 2025; Jing et al., 2025; Körner et al., 2026; Ravisankar et al., 2026).

Placement 2 will move from implementing these methods to applying them to a structured research problem: multilingual moral decision-making. The project will use the Moral Machine experiment as the basis for constructing a multilingual mechanistic benchmark (Awad et al., 2018). Moral Machine presents unavoidable autonomous-vehicle dilemmas in which a vehicle must either remain on its current path or swerve, with each action producing a different harmful outcome.

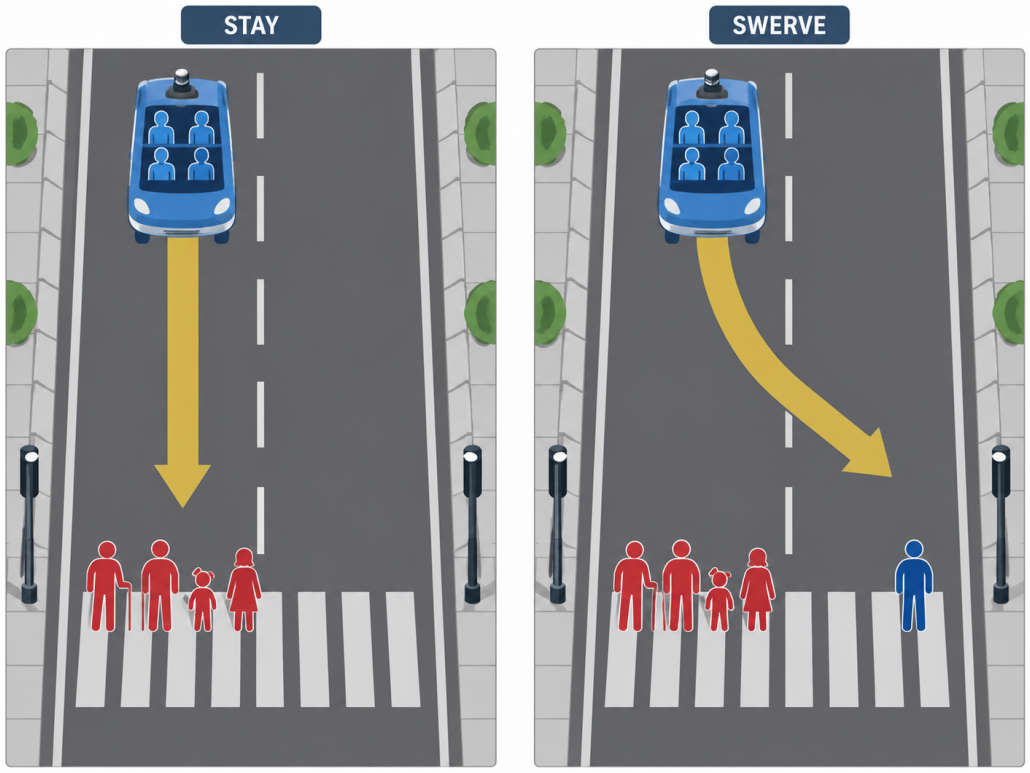



Figure 1: Illustrative Moral Machine-style dilemma. The vehicle must choose between remaining on its current path (`STAY`) or changing direction (`SWERVE`), with different consequences for each action.

The original Moral Machine experiment collected human decisions across many countries and measured preferences involving factors such as age, number of lives, species, social status, and road legality (Awad et al., 2018). These responses will not be treated as morally correct answers. Instead, they will provide descriptive human reference patterns against which LLM behaviour can be compared.

The benchmark will present semantically equivalent Moral Machine scenarios to LLMs in multiple languages and measure whether changing only the language changes the model's decision, confidence, or internal processing. Behavioural results will then be linked to mechanistic analysis using JLens, which will be used to examine how relevant verbalizable concepts develop across model layers and whether equivalent dilemmas produce similar or different internal concept trajectories across languages.

Selected scenarios showing strong cross-language differences will also be examined using targeted JLens interventions, such as strengthening, suppressing, or swapping selected concept directions. These interventions will test whether identified concepts contribute causally to changes in model behaviour.

The intended contribution is therefore a benchmark connecting human Moral Machine response patterns, multilingual LLM decisions, internal concept representations, and targeted causal intervention. Placement 2 extends the methodological work completed in Placement 1 into a broader empirical investigation of how multilingual LLMs represent and resolve moral trade-offs.

# 1 Proposed Activities

The placement will be an entirely computational research project using public research data, open-source large language models, published interpretability implementations, and university or research-group computing resources.

The main research question is:

> **How do multilingual LLMs differ in their behavioural and internal representations of moral trade-offs, and how do these differences relate to human cross-cultural preference patterns?**

A secondary question is:

> **Can targeted concept-level interventions change language-dependent moral decisions observed in the benchmark?**

## 1.1 Moral Machine Benchmark Construction

The benchmark will be constructed from the public Moral Machine dataset and associated experimental design (Awad et al., 2018). The original analysis measures nine preference dimensions. Six describe the characters involved in a dilemma: Gender, Fitness, Social Status, Age, Number of Characters, and Species. Three describe the structure of the driving situation: Intervention (stay versus swerve), Relation to the Autonomous Vehicle (passengers versus pedestrians), and Law (legal versus illegal crossing).

Each benchmark session will contain 13 binary-choice scenarios, following the structure of a completed Moral Machine session. The six character-focused dimensions will each appear twice, producing 12 scenarios, followed by one Random scenario. The three structural factors will vary within these scenarios rather than being treated as separate outputs.

For example, an Age scenario could be presented to the model as:

> *A self-driving car has suffered complete brake failure and must choose between two unavoidable outcomes. If the car stays on its current path, three elderly pedestrians will be harmed. If the car swerves, three young pedestrians will be harmed. What should the car do? Respond with only* `STAY` *or* `SWERVE`*.*

The same underlying scenario will then be presented in multiple languages while preserving the characters, group sizes, legality, relation to the vehicle, and action structure. This allows the language of the prompt to change while the moral decision being evaluated remains the same.

The human side of the benchmark will use country-level preference patterns derived from the released Moral Machine data. After restricting the data to countries with more than 100 users

who completed a full session, the current dataset contains 117 eligible countries. Rather than comparing countries using raw numbers of individual responses, each country will contribute one preference direction for each Moral Machine dimension. This prevents countries with very large respondent samples from dominating the cross-country comparison.

Countries will then be grouped according to the benchmark language associated with them, and the country-level preference directions within each language group will be aggregated to form a language-associated human reference profile. For a benchmark scenario, the relevant human preference direction can then be mapped to `STAY` or `SWERVE` according to how that scenario is constructed. Country and language will remain conceptually distinct; the language grouping is an aggregation strategy and does not imply that every respondent from a country used or speaks the assigned language.

An English master version of each scenario will first be created and checked for consistency. Machine translation, such as Google Translate, may then be used to generate semantically aligned versions in a broader set of languages represented across the 117 countries. Duplicate languages will not require separate translations for every country. Translation checks will focus on preserving the decision structure, character types, group sizes, legality, passenger/pedestrian relation, and the meaning of `STAY` and `SWERVE`. A manually reviewed subset will be used to check semantic equivalence before full inference.

The benchmark will use one 13-scenario session structure, matching a completed Moral Machine session. Each of the 13 scenarios will be translated into every selected benchmark language. If $L$ languages are included, this will produce $13 \times L$ prompt instances per model.

Behavioural evaluation will be performed across the full multilingual benchmark. The more computationally intensive JLens analysis will then focus on a selected subset of scenarios and languages showing particularly strong cross-language agreement or divergence. This allows broad behavioural coverage while keeping the deeper mechanistic analysis feasible within the placement workload.

## 1.2 Behavioural Evaluation

For each model, scenario, and language, I will record the final `STAY` or `SWERVE` decision and, where technically available, the probability or logit margin associated with each option. The behavioural analysis will examine how consistently the model makes the same decision across semantically equivalent languages, how its confidence changes between languages, and how different the full output distributions are. This will include measures such as cross-language prediction agreement, differences in $p(\texttt{STAY})$ and $p(\texttt{SWERVE})$, logit-margin differences, and Jensen–Shannon divergence or a comparable distributional measure.

The model's response for each language will also be compared with the corresponding language-associated human reference profile. For each benchmark scenario, the relevant human preference

direction will be mapped to the `STAY` or `SWERVE` option represented by that scenario. Human–LLM agreement will then be measured as the proportion of scenarios for which the model's choice is consistent with the corresponding human reference preference. Results will also be summarised by Moral Machine dimension to identify where model behaviour most closely matches or differs from the human reference.

The evaluation will therefore focus on consistency and divergence rather than moral correctness, identifying where model behaviour remains stable across languages, where it changes despite the underlying dilemma remaining the same, and how these patterns relate to the human response distributions used as reference points.

## 1.3 JLens Mechanistic Analysis

After the behavioural evaluation, selected scenarios will be analysed using the Jacobian Lens (JLens) (Gurnee et al., 2026). JLens will be used to identify verbalizable concepts that become represented across model layers as the model processes each dilemma. The goal is to test whether semantically equivalent scenarios in different languages lead to similar or different internal concept trajectories.

For example, in an age-based dilemma, concepts such as *young*, *elderly*, *child*, or *old* may become visible at different layers. By comparing these concept scores across languages, it becomes possible to examine whether the model reaches the same final `STAY` or `SWERVE` decision through similar or different internal representations. This allows the project to distinguish behavioural agreement from mechanistic agreement.

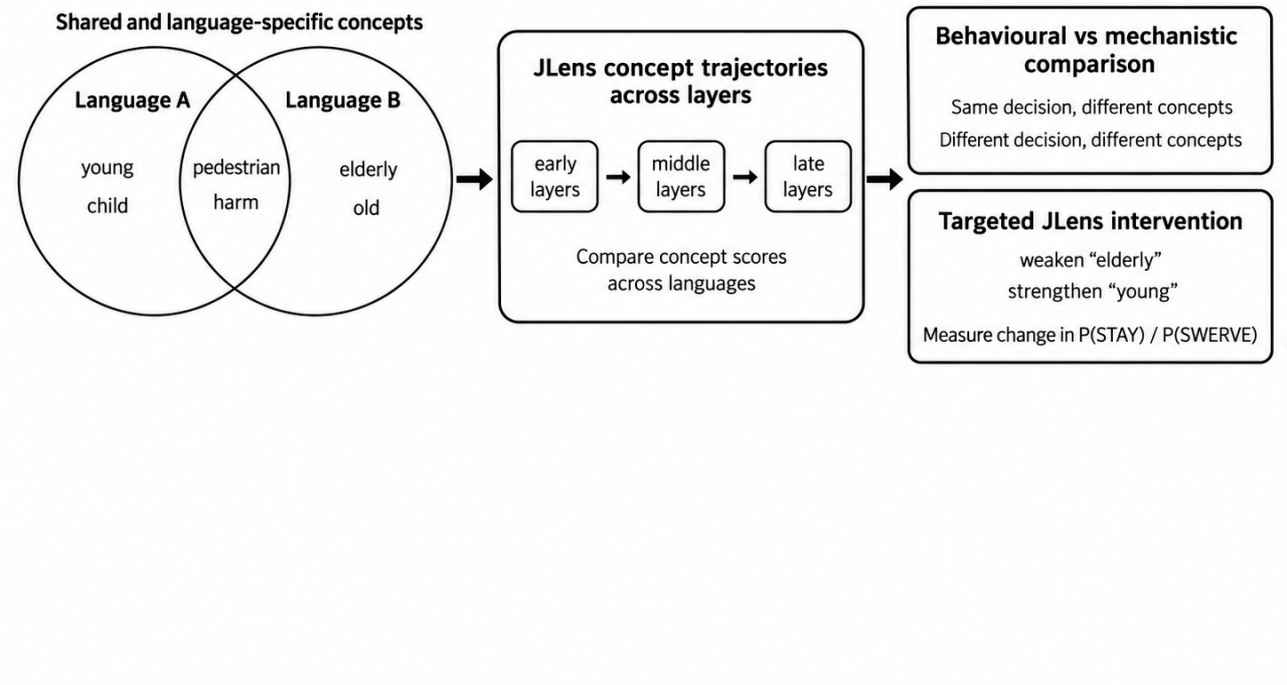



Figure 2: Illustrative JLens workflow for multilingual Moral Machine analysis. Shared and language-specific concepts are compared across model layers, then related to behavioural agreement or divergence and followed by targeted intervention.

As shown in Figure 2, the analysis will compare concept trajectories across languages and then interpret whether differences in those trajectories correspond to differences in model behaviour. Particular attention will be given to cases where the final decision differs across languages, as well as cases where the final decision is the same but the internal concept trajectories differ.

## 1.4 Targeted Concept Interventions

Whole-layer activation patching will not be a central Placement 2 method. Instead, the causal component of the project will focus on targeted concept-level intervention using JLens. Prior multilingual interpretability work has shown that changing internal representations can affect cross-lingual behaviour (Dumas et al., 2025; Ravisankar et al., 2026), and JLens provides a more focused way to test such effects at the concept level.

For a small number of high-divergence or otherwise informative scenarios, I will test interventions such as strengthening a selected concept direction, suppressing or ablating it, or swapping one concept direction for another. For example, in an Age dilemma, the effect of weakening an

*elderly*-associated direction or strengthening a *young*-associated direction can be evaluated by measuring the resulting change in $p(\texttt{STAY})$, $p(\texttt{SWERVE})$, or the final decision. This will provide causal evidence about whether the concepts identified by JLens contribute to language-dependent model behaviour, rather than treating concept readouts as descriptive evidence alone (Gurnee et al., 2026).

### 1.5 Models, Software, and Resources

The main experiments will use one open-source model in approximately the 7–9 billion parameter range. Gemma 2 9B is preferred because it was already used in Placement 1 and the existing interpretability workflow has been tested on it. If time and computational resources permit, a second model may be added to assess whether the main findings generalise across architectures.

Table 1: Main project resources.

| **Resource** | Use |
|---|---|
| **Python / PyTorch** | Model inference, activation collection, intervention, and analysis |
| **Hugging Face Transformers** | Loading open-source language models |
| **Moral Machine public data** | Scenario design and human-response reference distributions |
| **JLens implementation** | Layer-wise concept analysis and targeted concept intervention |
| **GPU compute** | Model inference and mechanistic interpretability experiments |
| **Git / GitHub** | Version control, code management, and reproducibility records |
| **University-approved storage** | Storage of prompts, results, logs, and processed data |

## 2 Induction / Training Requirements

This project is entirely computational and does not involve laboratory work, fieldwork, hazardous materials, recruitment of human participants, or direct interaction with identifiable participants.

Before beginning the placement, I will confirm with my mentor whether any research-group or computing inductions are required and ensure that relevant Macquarie University WHS and research-integrity requirements are current. Dataset, model, and software licences will be checked before use or redistribution, and GPU or remote-computing resources will be accessed only through authorised accounts.

The project will use the released public Moral Machine dataset and derived aggregate statistics. I will not attempt to re-identify participants and will use only the fields required for the benchmark analysis. If the project later changes to involve restricted data, identifiable data, or new human-

participant research, the relevant ethics and governance requirements will be confirmed before that work begins. If required by university policy, I will complete the Assessment of Sanctions Questionnaire before commencing the relevant research activity.

# 3 Proposed Timeline

The project is planned around approximately 35 hours of research activity, 7 hours of mentor meetings, and 20 hours of placement-deliverable preparation.

Table 2: Proposed week-by-week schedule.

| Week | Main activity | Research | Mentor | Deliverable |
|---|---|---|---|---|
| 1 | Finalise scope, benchmark design, selected languages, model, and evaluation protocol | 4 h | 1 h | 1 h |
| 2 | Construct the 13-scenario Moral Machine benchmark and validate multilingual prompt translations | 5 h | 1 h | 2 h |
| 3 | Run multilingual model inference and behavioural evaluation | 5 h | 1 h | 2 h |
| 4 | Prepare human-response reference profiles and compare them with model preference patterns | 5 h | 1 h | 2 h |
| 5 | Run JLens analysis on representative and high-divergence scenarios | 6 h | 1 h | 3 h |
| 6 | Compare cross-language concept trajectories and conduct targeted JLens interventions | 6 h | 1 h | 4 h |
| 7 | Integrate results and prepare final figures, report, presentation, and reproducibility material | 4 h | 1 h | 6 h |
| **Total** | | **35 h** | **7 h** | **20 h** |

The project will prioritise completing the multilingual benchmark and producing a clear behavioural and mechanistic analysis within the available placement time. If computational requirements are higher than expected, the number of languages or targeted JLens intervention cases will be reduced before increasing the overall scope. The minimum viable project will consist of one open-source model, the 13-scenario Moral Machine benchmark translated into a manageable set of languages, behavioural evaluation across those languages, JLens analysis of selected scenarios, and a small number of targeted concept-level interventions. If time and computational resources permit, additional languages or a second model may be included as extensions.

# 4 Project Data / Records / Creation

The project will produce several research outputs, including the multilingual benchmark prompts and translations, LLM decisions and probability records, country-level preference profiles, language-associated human reference profiles, JLens concept readouts, targeted intervention results, analysis code, figures, tables, and an experimental research log. Together, these materials will document both the behavioural and mechanistic stages of the project.

## Data Management Plan

Code, benchmark definitions, and documentation will be managed using Git and GitHub. Project files will be organised into separate directories for source-data references, benchmark prompts, translations, model outputs, human-reference summaries, JLens results, processed analyses, figures, tables, and research records. Raw public data will be kept separate from derived benchmark files and processed summaries so that the analysis pipeline remains clear and reproducible.

For the human-response analysis, I will record the filtering criteria used to select eligible countries, the number of respondents retained for each country, the country-level preference direction for each Moral Machine dimension, the language grouping assigned to each country, and the resulting language-associated human reference profiles. This will make the construction of the human reference data transparent and reproducible.

For each model experiment, I will record the model identifier and revision, software environment, benchmark version, language, decoding configuration, relevant layer and token positions, JLens settings, intervention parameters, date, and output location. This will allow final figures and reported results to be traced back to the exact experiment that produced them.

The full Moral Machine dataset is large, so the working project directory will contain only the subsets and aggregate outputs required for the placement. The original dataset source, version, and filtering procedure will be documented rather than duplicating unnecessary source files. Model weights will not normally be stored in the repository; instead, the exact model identifier and source will be recorded so that the experiments can be reproduced.

Benchmark prompts and multilingual translations will be version controlled so that changes in wording can be tracked. Successful, unsuccessful, and modified experiments will also be recorded in the research log. Final code, benchmark definitions, human-reference summaries, processed results, figures, and supporting documentation will be archived using approved university or research-group storage and backed up where appropriate.

# 5 Project Deliverable

For the FOSE7901 half-session placement, I propose the following division of the 40% placement deliverable:

| Component | Proposed weighting |
|---|---|
| Research presentation | 10% |
| Research records / reproducibility documentation | 5% |
| Final research report | 25% |
| **Total** | **40%** |

The research presentation will communicate the motivation for the benchmark, the Moral Machine scenario structure, the multilingual evaluation design, the human-reference aggregation method, and the main behavioural and JLens findings.

The research records will include benchmark-generation code, prompt and translation versions, model and dataset information, country-level and language-associated human reference summaries, experimental configurations, JLens settings, intervention parameters, analysis scripts, and the research log required to understand and reproduce the reported experiments.

The final report will describe the benchmark construction, human-response aggregation, multilingual behavioural evaluation, JLens concept analysis, targeted concept interventions, main findings, limitations, and recommendations for future work. The intended research output is a benchmark-style study that links human response patterns, multilingual LLM behaviour, internal concept representations, and targeted causal intervention, providing an empirical foundation for later publication or methods-focused mechanistic interpretability research.

# 6 Potential Examiners

Potential examiners will be discussed with my mentor before submission.

**Examiner 1:** [Name], [School / Department]

**Examiner 2:** [Name], [School / Department]

# References

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